```
// C++ code
//
const int led= 11; //pin value for led
const int trig= 6; //pin value for trig
const int echo= 5; //pin value for echo
long duration;
float distance;

void setup()
{
  pinMode(led, OUTPUT);
  pinMode(trig, OUTPUT);
  pinMode(echo, INPUT);
  Serial.begin(9600);
}

//custom map function

float mapf(float input, float ir1, float ir2, float or1, float or2)
{
  float diffi=input-ir1; //to see the difference between the input value and the start of the range
  float outrange= or2-or1; //to get the range of the output
  float inrange= ir2-ir1; //to get the range of the input
  float m= (diffi*outrange/inrange)+or1; //added or1 to get the value in the range of the output
//the final value is converted to output range so that the connected device can actually use the
value
  return m; //to send value back when the function is called
}

  void loop()
{
  digitalWrite(trig, LOW);
  delayMicroseconds(2);
  digitalWrite(trig, HIGH);
  delayMicroseconds(10);
  digitalWrite(trig,LOW);

  duration= pulseIn(echo, HIGH);
  distance= 0.034/2*duration;

  if(distance>=20&&distance<=200)
  {
  float m= mapf(distance,20.0,200.0,0.0,255.0);
```

```
  analogWrite(led, m);//analogwrite is used as the value will be in a range and not just high and
low
  Serial.print("Distance Range: \n 20 cm ---------- ");
  Serial.println(distance);
  Serial.print("LED Brightness: \n 0---------- ");
  Serial.println(m);
  }
  else
  {
    digitalWrite(led,LOW);//no light will be seen
    Serial.println("Outside the sensor range");//as the specified range is b/w 20 and 200
  }
}
```